Reference data available for each dataset

File: reference_data_by_dataset.tex · Git: Michael G. Prisant 4c7b2f1 · Txd: 11/12/26 with local modifications!

"Lab" means the Richardson Lab's own files; "ours" means what our pipeline uses now; "not located" means a source document says the files exist but they have not been found on disk. Pipeline diagram: `pipeline_block_diagram.pdf`.

Dataset	Chain list	Lab's coordinate files (as the lab used them)	Our coordinate files	Lab's filter output	Lab's filter rules	Lab's reference results
top100	Reconstructed from the lab's `top100H` files (100 entries, 108 chains)	100 files with hydrogens, Reduce 2.10.2: `~/Desktop/MyProgs/PolarWork/top100H/`, identical to GitHub `rlabduke/reference_data`	RCSB downloads, Reduce 4.16	None	Word et al. 1999b: atoms with $B \geq 40$ omitted (our filter: mainchain $B \leq 40$)	None located
top500	Reconstructed from the lab's kinemage-server list (494 entries, 523 chains)	500 files with hydrogens, Reduce 2.6.3–2.13.2: GitHub `rlabduke/reference_data` (2026-11-12)	RCSB downloads, Reduce 4.16	Filtered residues in `srcdata` (mainchain $B < 30$), 2004 and 2005	Lovell et al. 2003: "residues with $B < 30$"	Ramachandran `srcdata` and contour tables, 2004 and 2005: GitHub `rlabduke/reference_data`; the 2006 ϕ/ψ archive (`top500_issues.md`): not located
top8000	`Top8000-best_hom70_pdb_chain_list.csv` (7,737 entries, 7,957 chains)	2011 chain files, Reduce 3.15.091106: `~/Desktop/Data/top8000_chains_70_2011/` (sha256 4a6bb1b9…)	RCSB downloads, Reduce 4.16	The filtered residue lists themselves (`Top8000_Rama_2013.zip`); no masks	`get_srcdata.py` in `Top8000_Rama_2013.zip`	Filtered residue lists (`Top8000_Rama_2013.zip`); `.ndft` tables
top2018_full	Top2018 all-atom chain list v2.0 (Zenodo 10.5281/zenodo.5773255)	Pruned single-chain files (`*_pruned_all.pdb`, kept residues only, Reduce 3.3 hinge flips); whole-entry files the lab started from: not held	RCSB downloads, Reduce 3.3.160422 (4.16 for 4oan, 4uzg)	Pruned files: our `.mask` files are derived from them	Williams et al. 2022; Top2018 database tables `allatom_filters`, `mc_filters`	Top2018 database: `~/Desktop/Data/williams_top2018_sqlite/top2018_build.db`
top2018_mc	Top2018 mainchain chain list v2.01 (Zenodo 10.5281/zenodo.5777651)	Pruned single-chain files (`*_pruned_mc.pdb`); whole-entry files: not held	RCSB downloads, Reduce 4.16	Pruned files: our `.mask` files are derived from them	Williams et al. 2022; Top2018 database table `mc_filters`	Top2018 database (as above)

Masks for top2018. The lab did not supply mask files; it supplied the pruned chain files. `build_ersatz_pdbs.py` derives our `.mask` file for each chain from the residues present in the lab's pruned file. The filter (`filter_pruned_residues.py`) reads the file's `USER INC` records as the fragment boundaries. So for top2018 the lab's filter is already applied; we do not apply filter rules ourselves. The rules (Williams et al. 2022, and the database's per-residue filter flags) serve only to check the result.